Son Nguyen

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SUMMARY

Software Engineer with hands-on experience researching, developing, and deploying innovative backend systems and low-level infrastructure in both industry and research settings. Researcher focused on leveraging the latest technologies and frameworks to increase the performance and reliability of massively scaled cloud systems, data centers, and embedded/edge environments.

EDUCATION

- PhD in Computer Science and Engineering - 3.9 GPA, University of California at Santa Cruz — September 2025 - Present
- BS in Computer Science - 3.8 GPA, The George Washington University — August 2021 - May 2025

SKILLS

- **Engineering:** Systems Design, Backend & Infrastructure, Cloud Architecture, Microservices, Embedded Systems, Operating Systems, API Design, RESTful Web Services, Databases, DevOps, Testing
- **Technologies:** Python, C, C++, Rust, Java, JavaScript, Typescript, MongoDB, AWS, Linux Kernel, Kubernetes

RELEVANT EXPERIENCE

Graduate Researcher - LiteCast Research @ UCSC Sustainable Systems Lab — **January 2026 - July 2026**

- Created a lightweight method that predicts carbon emissions of data centers in each energy production region and enables the scheduling of jobs during low-carbon energy periods to minimize carbon emissions using the SARIMAX model.
- Achieved up to 90% carbon optimality for data centers while using 52x less data than state-of-the-art methods without requiring an accuracy comparison during forecasting.
- Adapted control theory principles to optimize the SARIMAX model's performance, increasing average optimality results from 85% to 90%.
- Compared the SARIMAX model's accuracy and concordance with training data to EnsembleCI, CarbonCast, and LightBGM, demonstrating that Litecast can achieve better results with less data by leveraging SARIMAX's exogenous variables.

Researcher - LibGLP-1 Research @ GW Systems Lab — **March 2025 - August 2025**

- Enabled binary patching for object files by extending the ELFIO library to support in-place section creation.
- Executed binary patching by parsing and splitting ELF binaries into data and text sections for seamless removal.
- Debloated software by applying static analysis to determine and remove unused sections, decreasing memory footprint by up to 70% on Composite and reducing total on-disk file sizes by over 25%.
- Guaranteed that only unused sections are eliminated by validating popular libraries such as openSSL and openCV.

Fullstack Engineer Intern @ XSTAR Fund Management LLC — **February 2025 - April 2025**

- Identified, troubleshot, and resolved frontend bugs for a portfolio investment website, including issues with form inputs, SSN/ITIN formatting, missing fields, broken error handling, and UI updates, to optimize the user experience.
- Fixed data-handling defects impacting data integrity by adding input sanitization and validation prior to database writes.

Undergraduate Researcher - Composer Research @ GW Systems Lab — **July 2024 - November 2024**

- Compared Composer and CAmkES to highlight CAmkES's lack of explicit virtual resource allocation at the user level.
- Updated, integrated, and evaluated Patina (an RTOS API) with Composer, demonstrating that Composer incur negligible overhead to the initialization time of core components, and microseconds overhead for service-level components.
- Demonstrated that Composer can shrink the capability footprint by over than 65%, reducing possible attack surfaces.

Undergraduate Researcher @ GW Systems Lab — **January 2024 - May 2024**

- Developed a tracing library/debugging tool in C to log & retrieve events in support of OS Research.
- Minimized overhead and ensured event logging is achieved in under 100 CPU cycles by implementing lockless data structures, compiled time argument counts, and function inlining.

Backend Engineer Intern @ Sciquel — **June 2022 - May 2023**

- Designed and implemented REST APIs for the SciQuel website that publishes and enables users to browse summaries of academic research to support security features, such as user authentication and password recovery, using Express.
- Maintained the organization's MongoDB database of rewritten science articles and metadata, such as publish timestamps, draft timestamps, author information, source references, tags, and editorial status, by identifying and resolving missing data and inconsistencies to maximize reliability and accuracy via automated scripts and schema enforcement.

Team Leader @ Design Build Fly's Testing Team, The George Washington University — **September 2022 - April 2023**

- Led the development of an embedded system on Arduino that accumulated flight data, such as flight and wind speed, pressure, and location, in collaboration with a team of 4 students.
- Visualized flight data by plotting graphs to model flight speed and display acceleration on G-meters using Python Plotly, and analyzed each flight to detect parts requiring rebuilding and identify possible improvement points.
- Led workshops to provide training on data analysis, visualization, and development with Pandas, Matplotlib, and Arduino.